

# Hyunseung Ryu

Hi, I'm Hyunseung Ryu. I'm a graduate of Gwangju Science Academy for the Gifted, now studying EECS and AI Convergence at GIST. Physics, math, astronomy, and problems you solve through simulation have held my attention for a long time; over the past two years I've become someone who works across frontend, backend, apps, and design.

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## Experience

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### Audio Intelligence Technology & Research Lab (AiTeR), School of EECS, GIST

2025.09 – Present

Frontend developer intern

Taking part in experiments and research at the audio intelligence lab.

[aiter.gist.ac.kr/](http://aiter.gist.ac.kr/) ↗

## Education

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### Gwangju Institute of Science and Technology (GIST)

2025.03 – Present

B.S., Electrical Engineering and Computer Science; AI Convergence (double major)

National Science and Engineering Scholarship, Korea Student Aid Foundation

Freshman representative, 2025 matriculation ceremony

Class representative, Electrical Engineering and Computer Science, 2026

Dean's List, Spring 2026 (GPA 4.406 / 4.5)

### Gwangju Science Academy for the Gifted

2022.03 – 2025.02

Natural Sciences

Computer science GPA 4.285 / 4.3

Gold Prize in every in-school computer science competition entered (2022, 2023), the 2022 one was the only award given to a first-year student

Six Subject Excellence Awards in computer science, four of them ranked 1st

## Selected projects

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### rhseung.me

2026.08

A personal site built on Astro islands. This page is the result.

### Campass

2024.11 – 2024.11

A companion service that gathers everything you need for campus life in one place.

### Table-tennis robot

2024.10 – 2025.01

Computer vision tracks the ball in real time and predicts the strike point to drive a four-axis linear robot.

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|----------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>NEAT</b>                                                                                                                      | 2024.09           |
| Evolving network topology itself with NEAT (NeuroEvolution of Augmenting Topologies).                                            |                   |
| <b>Glance</b>                                                                                                                    | 2024.06           |
| A Minecraft interface mod bundling a set of quality-of-life tweaks.                                                              |                   |
| <b>Fliggle</b>                                                                                                                   | 2024.05 – 2024.07 |
| A social app built for practice, with both the frontend and the backend written from scratch.                                    |                   |
| <b>IdP</b>                                                                                                                       | 2024.03 – 2024.08 |
| GIST's single sign-on system. OAuth 2.0 and OpenID Connect let one account log into every campus service, and external ones too. |                   |
| <b>Particles</b>                                                                                                                 | 2024.03 – 2024.11 |
| A constraint-based 2D physics engine. Position-based dynamics solve rigid bodies and joints.                                     |                   |
| <b>Pattern</b>                                                                                                                   | 2024.02 – 2024.05 |
| Builds regular expressions through a builder API — method chains instead of unreadable expressions.                              |                   |
| <b>DateTime</b>                                                                                                                  | 2024.01 – 2024.04 |
| Parses natural-language expressions like "next Tuesday" into dates.                                                              |                   |
| <b>siunits</b>                                                                                                                   | 2023.12           |
| A unit-arithmetic library for physics experiments. Dimensionally invalid operations fail at the type level.                      |                   |
| <b>GSA KakaoTalk bot</b>                                                                                                         | 2023.09 – 2024.02 |
| A KakaoTalk bot for Gwangju Science Academy — meals, timetables and the academic calendar without leaving the chat.              |                   |
| <b>YOLOv3 from scratch</b>                                                                                                       | 2023.03 – 2023.11 |
| Reimplemented YOLOv3 from the paper alone — anchor boxes and multi-scale detection included.                                     |                   |
| <b>Competitions</b>                                                                                                              |                   |
| <hr/>                                                                                                                            |                   |
| <b>3rd place, 2026 AX Challenge Track 1</b>                                                                                      | 2026.08           |
| Table-tennis robot track · Jun – Aug 2026                                                                                        |                   |
| <b>Participant, AI Creative Convergence Competition</b>                                                                          | 2025              |
| GIST AI Graduate School                                                                                                          |                   |
| Table-tennis robot track                                                                                                         |                   |
| <b>Participant, JUNCTION ASIA 2025</b>                                                                                           | 2025              |
| ENTBE                                                                                                                            |                   |
| <b>Participant, L:AUNCH AI Career School</b>                                                                                     | 2025              |
| Root Impact · Krypton X                                                                                                          |                   |
| Supported by Google.org                                                                                                          |                   |

# Winner, 2025 Honam Regional Mini Tex-Corps

2025

Ministry of Science and ICT · COMPA

Hosted by GIST, Ewha Womans University and Wonkwang University startup centers

## Skills

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### Languages

TypeScript JavaScript HTML5 CSS3 C C++ Rust Java Kotlin Swift Dart Python  
Shell LaTeX

### Web Frontend

React Next.js Astro TanStack Start TanStack Router TanStack Query Tailwind CSS  
styled-components Vite i18next

### Mobile Frontend

Flutter React Native SwiftUI

### Backend

Node.js Nest.js Flask Prisma PostgreSQL MySQL

### Tools

Git GitHub Actions Storybook Playwright Figma Slack Notion 1Password

### MLDL

PyTorch OpenCV NumPy Pandas Jupyter